ASSESSMENT OF PREDISPOSING FACTORS TO TREATMENT FAILURE AMONG TUBERCULOSIS PATIENTS ATTENDING TB CLINICS IN KWARA STATE

Final Report

By

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23/57ML/949

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Prepared Under the Grant From

Kwara State University Undergraduate Research Programme

Submitted To

The Center for Research and Development, Kwara State University, Malete, P.M.B. 1530, Ilorin, Kwara State, Nigeria

AUGUST 2026

DECLARATION

I hereby declare that this research titled **"Assessment of Predisposing Factors to Treatment Failure Among Tuberculosis Patients Attending TB Clinics in Kwara State"** was carried out by me under a grant from Kwara State University Undergraduate Research Programme during the 2025/2026 academic session. I also declare that the information provided herein are original and those that are not original are duly acknowledged.

Faridah Ayomide Uthman	Signature and Date

CERTIFICATION

This is to certify that the research titled **"Assessment of Predisposing Factors to Treatment Failure Among Tuberculosis Patients Attending TB Clinics in Kwara State"** was carried out by **Faridah Ayomide Uthman,** under a grant from Kwara State University Undergraduate Research Programme during the 2025/2026 academic session.

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ACKNOWLEDGEMENTS

All praise and gratitude are due to the Almighty Allah for His divine guidance, strength, and wisdom, which enabled me to successfully complete this project. His blessings sustained me throughout the course of this research work.

My profound gratitude goes to my project supervisor, Mr. Munirudeen Ibrahim, for his invaluable supervision, constructive criticism, patience, and continuous guidance throughout this research. His support and expertise were instrumental in the successful completion of this project.

I sincerely appreciate the support and leadership of the Head of Department, Mr. Abdurrazaq Mustapha, and to my esteemed lecturers in the like of Dr. Garuba O.W., Mr. Ahmad, Mr. Abdullahi and all the lecturers in the Department of Medical Laboratory Science for their academic guidance, encouragement, and support, which contributed greatly to the success of this project.

I am deeply grateful to my parents, Mr. and Mrs. Usman for their unwavering love, sacrifices, prayers, and moral and financial support. Their belief in me served as a major source of motivation during this academic pursuit. And to my siblings, Fathi, Fatihah, and Fareed for their love and belief in me.

I also extend my sincere appreciation to Sci. Kolo Joshua, Mrs. Rasheedat Ahmed, Miss Kaosarah Soliu, Muhammad Aiyelero, Mrs. Rasheedat Lukman, Mrs. Mariam Rufai for their assistant in data collection and support throughout the course of this work.

I also extend my sincere appreciation to Adedeji Abdulkabeer, Bature Muzzammil, Isah Ruqayyah, Ajibola Abdulmalik, Mudashir Jamiu and all individuals who contributed directly or indirectly to the success of this project. May Allah reward you all abundantly.

Finally, I acknowledge my own dedication, perseverance and commitment throughout the course of this research. I am grateful to Allah for granting me the strength, wisdom, patience, and resilience to see this project through to completion. I pray that the knowledge and experience gained from this study will contribute positively to my academic and professional growth. Ameen.

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ABSTRACT

Tuberculosis (TB) is an infectious disease caused by *Mycobacterium tuberculosis* and remains a major public-health problem, particularly in developing countries such as Nigeria. Despite the availability of effective anti-tuberculosis therapy, incomplete or irregular treatment has contributed to the emergence of drug-resistant forms of TB, including rifampicin-resistant tuberculosis (RR-TB), multidrug-resistant tuberculosis (MDR-TB), and extensively drug-resistant tuberculosis (XDR-TB), which pose a serious challenge to TB control programmes. This study was designed to determine the predisposing factors responsible for treatment failure and the emergence of drug-resistant tuberculosis among patients attending TB clinics in Kwara State. A descriptive cross-sectional, hospital-based design was adopted, involving a minimum of 261 adult patients (≥15 years) with confirmed TB diagnosis attending Tuberculosis Diagnostic Centers in Kwara State, recruited by consecutive sampling over a three-month period. Data was obtained through a structured data extraction form retrieving sociodemographic, clinical, and laboratory information from patient records, complemented by a structured interviewer-administered questionnaire covering treatment adherence, lifestyle, household and health-system factors. Data was analyzed using Microsoft Excel and SPSS version 26.0. Most of the participants were female (54.8%) and were within the age range of 20-29 years (42.5%) with less than ₦50,000 monthly income. The prevalence of drug-resistant tuberculosis (DR-TB) in this study was 4.2%. The significant predisposing factors to treatment failure among TB patients in this study include missing of TB medication for more than 2 consecutive days (16.8%), stopping of TB treatment before completion (7.4%), far distance to the treatment center (17.2%), history of contact with TB patients in the household (12.3%) and close contact with drug-resistant TB patients (12.3%). The identified predisposing factors associated with the emergence of DR-TB in this study include far distance to the treatment facility between 5-10km ($p<0.001$), close contact with drug-resistant tuberculosis patient ($p=0.018$) and history of contact with TB patients in household ($p=0.001$). The study identified geographical access, healthcare access and household-contact factors as important predisposing factors influencing tuberculosis treatment outcomes. Improving geographical access to treatment services, expanding insurance coverage, providing socio-economic support and strengthening adherence counselling may enhance treatment success and reduce the burden of drug-resistant tuberculosis in Kwara State.

CHAPTER ONE

1.0 INTRODUCTION

1.1 BACKGROUND OF STUDY

Tuberculosis (TB) is a chronic infectious disease caused by *Mycobacterium tuberculosis*, an acid-fast bacillus that primarily affects the lungs but is also capable of involving other organs of the body. It remains one of the leading causes of death from a single infectious agent globally, and its burden is disproportionately borne by low- and middle-income countries, including Nigeria (World Health Organization, 2025). Nigeria currently ranks among the countries with the highest burden of tuberculosis in the world, and the disease continues to exact a heavy toll on public health resources, productivity, and household income, particularly among the poorest segments of the population.

Although effective first-line and second-line anti-tuberculosis drug regimens exist, the successful treatment of TB depends heavily on strict, uninterrupted adherence to prescribed therapy over several months. Where this adherence is compromised, either through irregular drug intake, premature discontinuation of therapy, inadequate dosing, or poor-quality medication, the tubercle bacilli are exposed to sub-therapeutic drug concentrations. This creates selective pressure that favours the survival and proliferation of drug-resistant mutants, giving rise progressively to multidrug-resistant tuberculosis (MDR-TB), rifampicin-resistant tuberculosis (RR-TB), and ultimately extensively drug-resistant tuberculosis (XDR-TB) (Otchere *et al* ., 2024).

Rifampicin-resistant tuberculosis (RR-TB) refers to TB caused by strains of *Mycobacterium tuberculosis* that are resistant to rifampicin, detected with or without resistance to other anti-TB drugs, most identified using rapid molecular tests such as GeneXpert MTB/RIF assay (WHO,2023). Multidrug-resistant tuberculosis (MDR-TB) is defined as the TB resistant to at least both rifampicin and isoniazid, the two most effective first-line anti-tuberculosis drugs (WHO,2023). Extensively drug-resistant tuberculosis (XDR-TB) is defined as tuberculosis caused by strains of M. tuberculosis that are resistant to rifampicin and isoniazid, the two most potent first-line drugs, in addition to resistance to any fluoroquinolone and at least one of the second-line

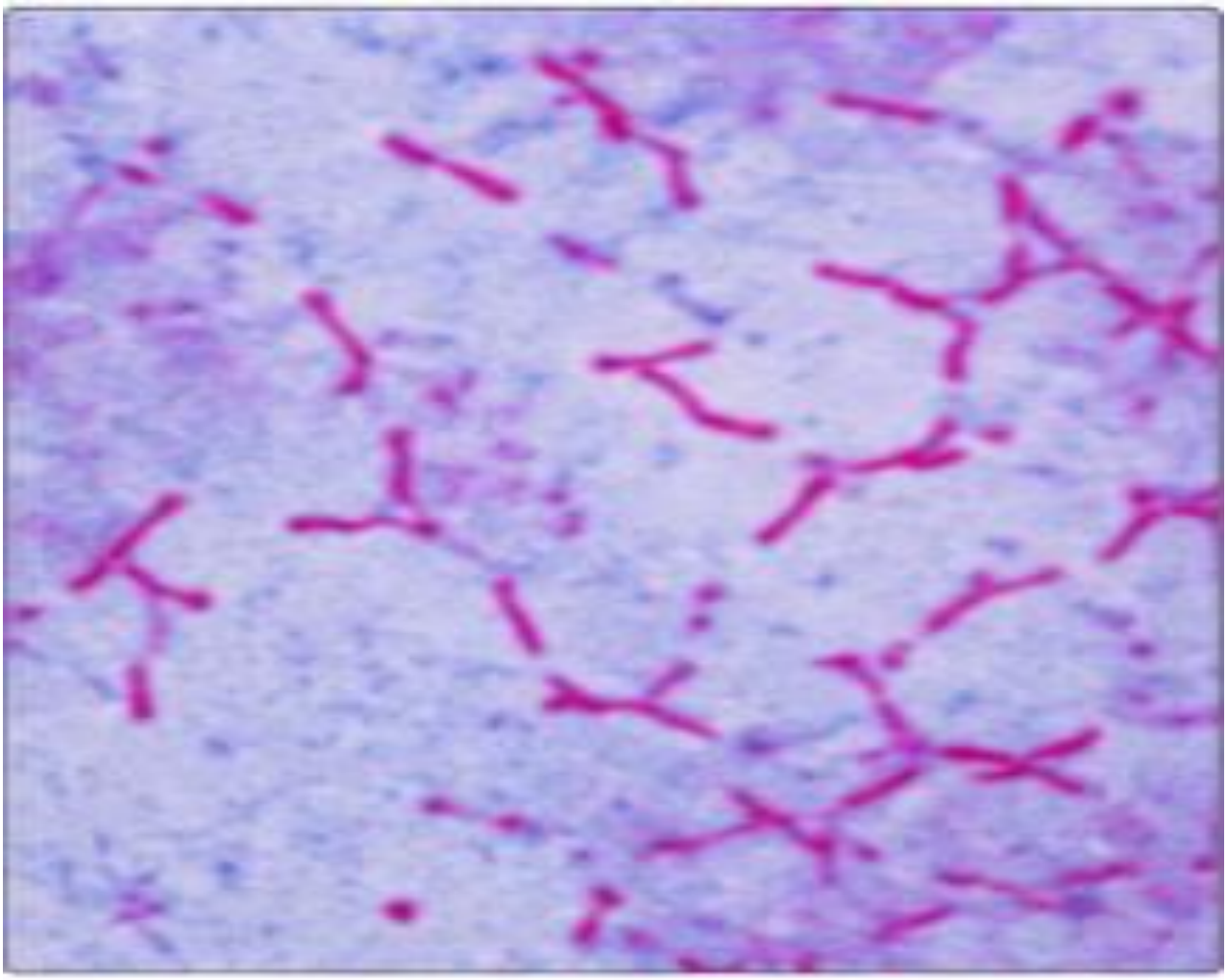


Figure 1.1: *Mycobacterium tuberculosis* stained by Ziehl-Neelsen method (WHO, Global tuberculosis report, 2025).

injectable agents such as kanamycin, amikacin, or capreomycin (WHO, 2023). Patients with XDR-TB have markedly fewer treatment options, must be placed on longer and more toxic regimens, and experience considerably poorer treatment outcomes, including higher rates of treatment failure and death, compared with patients who have drug-susceptible TB (Shehu *et al* ., 2025).

In Kwara State, as in many parts of Nigeria, TB clinics continue to record cases of treatment failure and drug resistance among patients who had previously been placed on anti-tuberculosis therapy. A range of predisposing factors have been implicated in this pattern, spanning socio-demographic characteristics, clinical history, treatment adherence behavior, lifestyle practices, household and environmental conditions, and health system-related barriers such as drug stock-outs and distance to treatment centers (Murphy-Okpala *et al* ., 2025). Understanding the relative contribution of these factors within the local context of Kwara State is essential for designing targeted interventions capable of curbing the emergence and spread of drug-resistant TB, including its most severe form, XDR-TB.

1.1.1 OVERVIEW OF TUBERCULOSIS AND DRUG RESISTANCE

Tuberculosis has remained a leading cause of death from a single infectious agent for much of the past decade. Nigeria has consistently featured among the countries with the highest estimated TB incidence and among those with a substantial gap between estimated and notified cases, meaning a considerable proportion of infectious cases go undiagnosed or are diagnosed late (World Health Organization, 2023). Drug resistance in M. tuberculosis develops principally through spontaneous chromosomal mutation, but the selection and amplification of resistant strains within a patient population is driven largely by human and health-system factors, chief among them inconsistent or incomplete drug exposure during treatment (Otchere *et al* ., 2024).

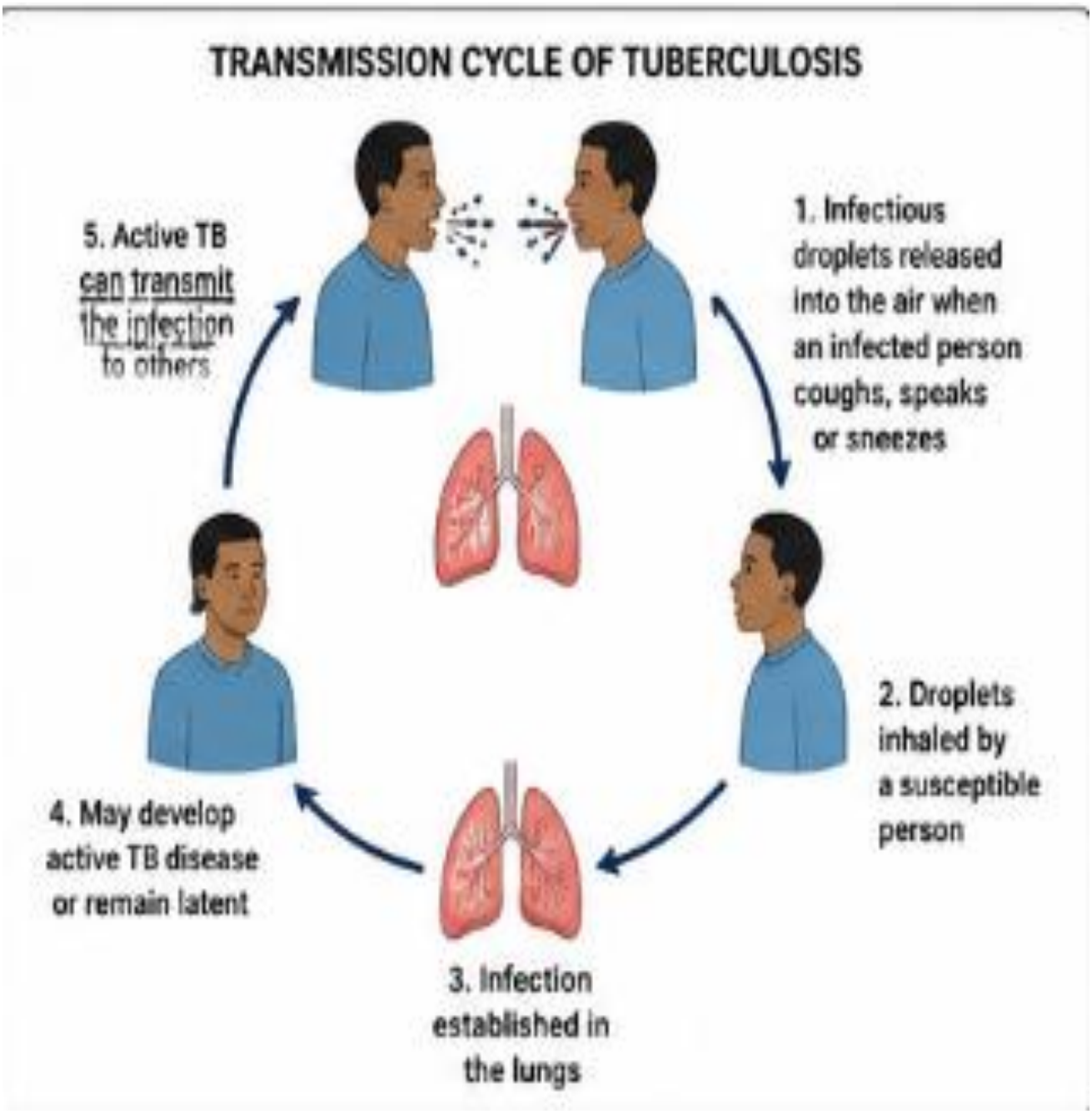


Figure 1.2: Transmission cycle of tuberculosis (WHO).

1.1.2 DRUG RESISTANT TUBERCULOSIS: GLOBAL AND NIGERIAN PERSPECTIVE

Rifampicin-resistant tuberculosis (RR-TB) refers to TB caused by strains of *Mycobacterium tuberculosis* that are resistant to rifampicin, detected with or without resistance to other anti-TB drugs, most identified using rapid molecular tests such as the GeneXpert MTB/RIF assay (WHO, 2023). Because rifampicin is one of the most potent and widely used first-line anti-TB drugs, resistance to it is treated clinically as a strong proxy for multidrug resistance, and the World Health Organization recommends that patients with confirmed RR-TB be placed on a multidrug-resistant TB treatment regimen even before full drug-susceptibility results are available. RR-TB therefore represents an important early warning point in the resistance pathway, and its timely detection is central to preventing progression to more extensive forms of resistance.

Multidrug-resistant tuberculosis (MDR-TB) is defined as TB resistant to at least both rifampicin and isoniazid, the two most effective first-line anti-tuberculosis drugs (WHO, 2023). MDR-TB typically develops when patients with drug-susceptible or mono-resistant TB are exposed to inadequate treatment, whether through incomplete drug regimens, poor adherence, substandard drug quality, or premature treatment discontinuation, allowing resistant mutants to survive selection pressure and become the dominant bacterial population (Otchere *et al* ., 2024). Once established, MDR-TB requires longer, costlier, and more toxic second-line treatment regimens, and patients face a substantially higher risk of poor treatment outcomes than those with drug-susceptible TB. In Nigeria, MDR-TB has been reported across several regions, with treatment-programme weaknesses such as inconsistent drug supply and inadequate follow-up identified as recurring contributors to its persistence (Shehu *et al* ., 2025). Left inadequately managed, MDR-TB can progress further along the resistance spectrum, first to pre-XDR-TB and ultimately to extensively drug-resistant tuberculosis (XDR-TB), which is discussed in the following section.

XDR-TB represents the most advanced and difficult-to-treat form of drug resistance encountered in routine TB care, arising as an extension of multidrug resistance once fluoroquinolone and second-line injectable resistance is acquired (WHO, 2023). Shehu *et al* . (2025) reviewed the status of drug-resistant tuberculosis in Nigeria and highlighted that treatment-programme weaknesses, including inconsistent drug supply, inadequate laboratory capacity for early resistance detection, and poor patient tracking, have contributed to the persistence and spread of resistant strains across

several states. A 2025 issue note from the BRICS Tuberculosis Research Network similarly identified gaps in diagnostic coverage and treatment support systems as recurring challenges across member countries with a high TB burden, Nigeria included (BRICS Tuberculosis Research Network, 2025). Regionally, Oladimeji *et al* . (2022), in a retrospective review of drug-resistant tuberculosis presentation patterns in Nigeria, reported a prevalence of approximately 19.18% for drug-resistant TB in the North Central zone, providing the basis for the sample size estimate adopted in this study.

1.1.3 PREDISPOSING FACTORS TO TREATMENT FAILURE AND DRUG-RESISTANCE

Several studies have attempted to characterize the factors that predispose TB patients to incomplete treatment and, subsequently, to drug resistance. Murphy-Okpala *et al* . (2025), evaluating decentralized drug-resistant TB services in Southern Nigeria, found that determinants of poor treatment outcomes clustered around pre-treatment loss-to-follow-up, weak multilevel referral systems, and socio-economic hardship affecting patients' ability to sustain clinic visits. Otchere *et al* . (2024) similarly reviewed factors associated with the development of drug-resistant tuberculosis and identified HIV co-infection, alcohol and substance use, overcrowded living conditions, and interruptions in drug supply as recurring contributors across multiple settings. Shehu *et al* . (2025) further noted that health-system-related barriers, such as delays in diagnosis, stock-outs of anti-TB medication, and limited access to directly observed therapy (DOT), compound patient-level adherence challenges, creating a combined effect that increases the risk of treatment failure.

Taken together, the literature suggests that predisposing factors to drug-resistant TB are multi-layered, spanning individual patient behavior, household and environmental circumstances, and the capacity of the health system to support consistent treatment delivery. Much of the specific Nigerian evidence located for this review, notably Murphy-Okpala *et al* . (2025), is drawn from Southern Nigeria, with comparatively little published data specific to Kwara State or the wider North Central region.

While the broad categories of predisposing factors to drug-resistant tuberculosis are well described in the Nigerian literature, there remains a paucity of locally generated data from Kwara State that

quantifies the relative contribution of socio-demographic, clinical, adherence-related, lifestyle, household, and health-system factors specifically among drug-resistant TB patients. This study seeks to address that gap by generating context-specific evidence from TB clinics within the state.

1.2 STATEMENT OF PROBLEM

Tuberculosis remains a major public-health challenge in Nigeria despite decades of national and international control efforts and the ready availability of effective standardized treatment regimens. The emergence of drug-resistant forms of TB, including rifampicin-resistant tuberculosis (RR-TB), multidrug-resistant tuberculosis (MDR-TB), and extensively drug-resistant tuberculosis (XDR-TB), has complicated this picture considerably, as it renders many of the country's standard treatment protocols ineffective and demands second-line regimens that are more expensive, more toxic, and considerably longer in duration.

A recurring observation in TB clinics is that a substantial proportion of patients who go on to develop drug-resistant TB have a history of incomplete, irregular, or interrupted prior treatment. Reasons commonly cited include adverse drug reactions, financial hardship, inadequate patient education, poor follow-up systems, long distances to treatment centers, and occasional stock-outs of anti-TB medication. However, the precise pattern and relative weight of these predisposing factors within Kwara State have not been sufficiently documented, making it difficult for local health authorities to priorities interventions. Without empirical, context-specific evidence on what drives incomplete treatment and subsequent drug resistance, TB control programmes in the state risk applying generic solutions that do not address the actual drivers of the problem, allowing the burden of drug-resistant TB to persist or worsen.

1.3 JUSTIFICATION OF THE STUDY

Incomplete tuberculosis treatment remains one of the most significant and modifiable contributors to the emergence of drug-resistant tuberculosis. Unlike some risk factors that are difficult to alter, factors related to treatment adherence, patient education, and health-system support are, in principle, amenable to intervention once they are properly identified and understood by systematically assessing the socio-demographic, clinical, behavioral, and health-system factors associated with treatment failure among drug-resistant TB patients attending TB clinics in Kwara State, this study generates locally relevant evidence that can inform the design of targeted

adherence-support programmes, strengthen directly observed therapy (DOT) implementation, and guide the allocation of resources toward the specific barriers patients face.

The findings are expected to be of direct value to clinicians managing TB patients, laboratory scientists involved in resistance surveillance, hospital administrators, and public-health policymakers responsible for tuberculosis control programming in Kwara State and, by extension, similar settings across Nigeria.

This study contributes directly to Sustainable Development Goal 3 (Good Health and Well-Being), and specifically to Target 3.3, which commits to ending the epidemics of tuberculosis and other communicable diseases by 2030. By identifying the factors responsible for treatment failure among drug-resistant TB patients, the study generates evidence that can strengthen national and state-level TB control programmes, improve treatment adherence monitoring, and ultimately contribute to reducing the transmission of drug-resistant tuberculosis strains.

1.4 AIM OF THE STUDY

To determine the predisposing factors responsible for treatment failure and the emergence of drug-resistant tuberculosis among patients attending TB clinics in Kwara State.

1.5 OBJECTIVES OF THE STUDY

1. To determine the prevalence of drug-resistant tuberculosis (RR-TB, MDR-TB, and XDR-TB) among previously treated tuberculosis patients attending TB clinics in Kwara State.

2. To identify the predisposing factors responsible for incomplete or interrupted anti-tuberculosis treatment among the study population.

3. To determine the association between the identified predisposing factors and the emergence of drug-resistant TB among the study participants.

1.6 RESEARCH HYPOTHESES

Null Hypothesis (H_0): There is no statistically significant association between selected risk factors and the emergence of drug-resistant tuberculosis among tuberculosis patients attending TB clinics in Kwara State.

Alternative Hypothesis (H_1): There is a statistically significant association between selected risk factors and the emergence of drug-resistant tuberculosis among tuberculosis patients attending TB clinics in Kwara State.

1.7 SIGNIFICANCE OF THE STUDY

This study provides baseline, evidence-based data on the burden and predisposing factors of drug-resistant TB within TB clinics in Kwara State, an area where locally specific data remain limited. The findings assist healthcare providers in identifying patients at higher risk of treatment failure early enough for closer monitoring and support. For policymakers and TB control programme managers, the results offer a practical basis for redesigning adherence-support strategies, strengthening DOT supervision, and improving drug supply-chain management to prevent stock-outs. Academically, the study also adds to the relatively scarce body of literature on drug-resistant tuberculosis in North Central Nigeria and may serve as a reference point for future research in the region.

CHAPTER TWO

2.0 METHODOLOGY

2.1 MATERIALS

The following materials, reagents, and equipment were used in the course of this study:

• Structured questionnaire (Appendix I)

• Structured data extraction form

• Patients' case files and TB treatment registers

• Personal protective equipment (PPE): disposable gloves, N95 respirators/face masks, and laboratory coats

• Informed consent forms

• Stationery (pens, notebooks, data recording sheets, files)

• Computer system installed with Microsoft Excel and SPSS version 26.0 for data entry and statistical analysis

2.2 METHODS

2.2.1 STUDY DESIGN

This study adopts a descriptive cross-sectional, hospital-based design, which allows for the simultaneous assessment of exposure (predisposing factors) and outcome (drug-resistant TB status) at a single point in time among the study population, making it suitable for estimating prevalence and examining associations between selected risk factors and treatment failure.

2.2.2 STUDY AREA

The study was conducted at Tuberculosis Diagnostic Centers (Kwara State University Teaching Hospital, Kwara State Specialist Hospital and General Hospital Offa) in Kwara State. These

centers serve as the primary points of care for the diagnosis, treatment initiation, and follow-up of tuberculosis patients, including those with drug-resistant forms of the disease, within the state.

2.2.3 ETHICAL CONSIDERATION

Ethical approval for this study was obtained from the Health Research Ethics Committee of Kwara State University Teaching Hospital (KWASUTH), Ilorin, prior to commencement of data collection. (KWASUTH/IRC/246/VOL. III/14).

2.2.4 STUDY DURATION

The study lasts for three months, a period that encompasses data retrieval from patient records, questionnaire administration, laboratory data extraction, statistical analysis, and report writing.

2.2.5 SAMPLE SIZE DETERMINATION

The sample size for this study was determined using the single population proportion (Cochran's) formula for descriptive cross-sectional studies:

$$n_0 = \frac{Z^2 p\,(1-p)}{e^2}$$

Where:

n = minimum required sample size

Z = standard normal deviate corresponding to a 95% confidence level = 1.96

P = estimated prevalence of drug-resistant tuberculosis is 19.18% or 0.1918 (Oladimeji *et al* . 2022)

e = margin of error = 5% or 0.05

Substituting into the formula:

$$n = \frac{1.96^2 \times 0.1918 \times (1 - 0.1918)}{0.05^2}$$

$$n = \frac{3.8416 \times 0.1918 \times 0.8082}{0.0025}$$

n= 238.

To calculate 10% attrition,

n= Original number × 1.10

n= 238 × 1.10

n= 261

Therefore, the total sample size required for this study was 261 participants.

2.2.6 STUDY POPULATION

The study population comprised patients receiving tuberculosis treatment and attending TB clinics in Kwara State, Ilorin, with particular focus on those who have a documented history of previous anti-tuberculosis treatment and confirmed drug-resistant TB status. Based on the sample size determination, a total of 261 participants were recruited for the study.

2.2.7 INCLUSION CRITERIA

The following categories of patients were included in the study:

• Adult patients (≥15 years) with confirmed MDR-TB, RR-TB, or XDR-TB diagnosed by molecular or culture-based methods.

• Patients with complete treatment history and laboratory records available in TB registers or hospital records.

• Patients who are willing to participate and provide informed consent.

2.2.8 EXCLUSION CRITERIA

The following categories of patients were excluded from the study:

• Pediatric patients (<15 years).

• Patients with no or incomplete treatment history.

• Patients who decline to participate or withdraw consent.

2.2.9 INFORMED CONSENT

Written informed consent was obtained from each participant after the purpose, procedures, risks, and benefits of the study have been explained in a language they understand. Participation was entirely voluntary, and participants retain the right to withdraw at any stage without any effect on their ongoing clinical care. Confidentiality of participant information was maintained throughout the study, with data extraction forms and questionnaires identified by code rather than names, and all records stored securely and accessible only to the research team.

2.2.10 DATA EXTRACTION FORM

A data extraction form is a structured tool used to systematically collect relevant information from patient records, TB registers, laboratory records or questionnaires. It consisted of the following sections: socio-demographic information, clinical information, treatment adherence information, lifestyle and environmental factors, health system-related factors and outcome variable.

2.2.11 DATA COLLECTION

Two complementary methods were used to collect data for this study. First, a structured data extraction form was used to retrieve relevant sociodemographic, clinical, and laboratory information, including GeneXpert and drug-susceptibility testing results, from eligible patients' TB registers and hospital records. Second, a structured interviewer-administered questionnaire (Appendix I) was used to obtain additional information directly from consenting patients on socio-demographic and socio-economic characteristics, clinical history, treatment adherence and drug use behavior, lifestyle and risk factors, household and environmental conditions, and health-system and access-related factors. The questionnaire was administered by the trained research assistants in a private setting to encourage candid responses, particularly on sensitive items such as treatment interruption and substance use.

Prior to full-scale administration, the data extraction form and questionnaire were reviewed by the study supervisor for content validity and pretested on a small number of patients outside the main study area to identify and correct any ambiguities in wording or structure.

2.2.12 DATA ANALYSIS

Data generated from the GeneXpert results, drug-susceptibility testing records, and patient treatment histories was entered into Microsoft Excel and subsequently analyzed using the Statistical Package for Social Sciences (SPSS), version 26.0. Descriptive statistics, including frequencies, percentages, means, and standard deviations as appropriate, were used to summarize the sociodemographic characteristics of participants and to determine the proportion of drug-resistant TB (RR-TB, MDR-TB, and XDR-TB) among incompletely treated tuberculosis patients.

The relationship between incomplete treatment and the occurrence of drug resistance, as well as the association between other selected predisposing factors and drug-resistant TB status, was assessed using the Chi-square test of association. All statistical tests were conducted at a 95% confidence interval, with a p-value ≤ 0.05 considered statistically significant. Results of the analysis was presented using tables, charts, and graphs to facilitate clear interpretation of the findings.

CHAPTER THREE

3.0 RESULTS

The results of the study involving 261 tuberculosis patients were presented in tables (3.1-3.4) and figures (3.1-3.6).

Figure 3.1 presents 4.2% (11) prevalence rate of drug-resistant tuberculosis (DR-TB) among the participants studied.

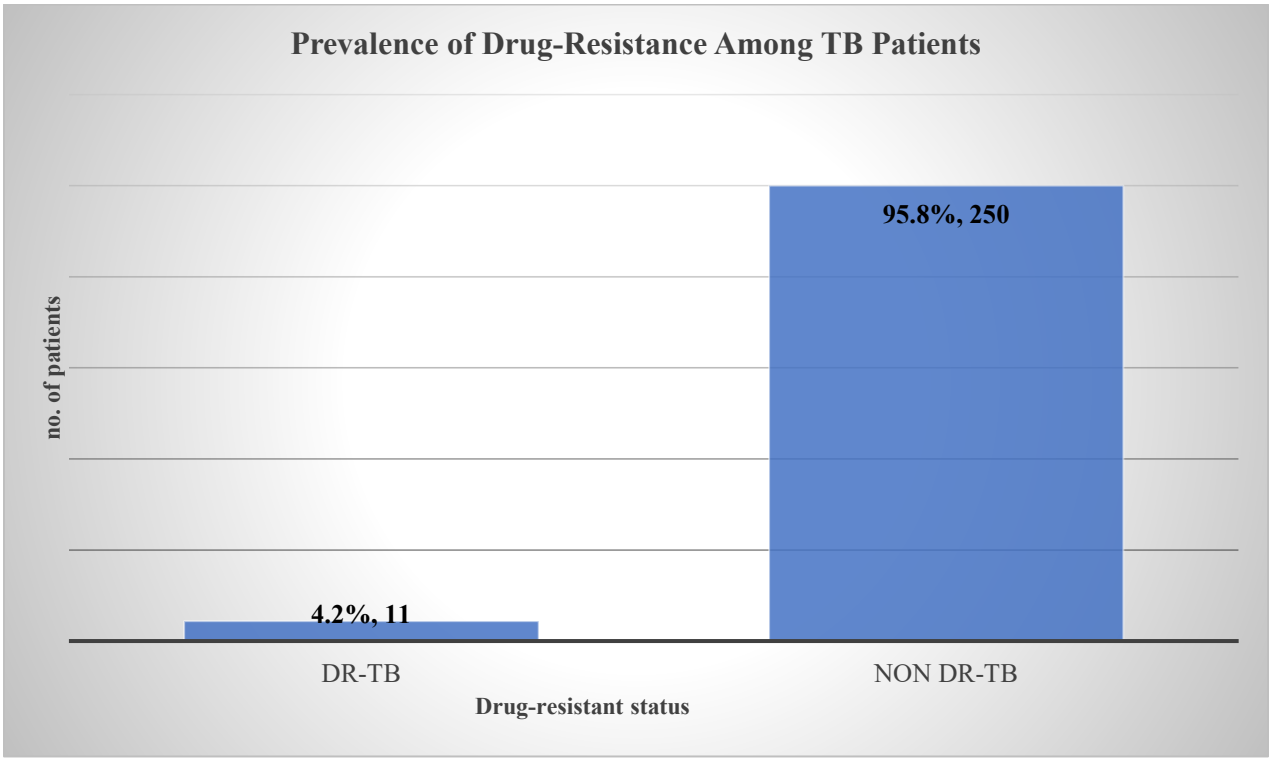


Figure 3.1: Prevalence of Drug-resistance among TB patients.

Table 3.1 shows the socio-demographic characteristics of the study participants. The age group 20-29 years had the highest frequency, accounting for 111 (42.5%) of the participants. Most of the participants 143 (54.8%) were female and the largest proportion of them were married 130 (49.8%). In addition, participants employed in an informal mode of occupation constituted the largest proportion of 114 (44.1%).

Table 3.1: Socio-Demographic Characteristics

Variables	Frequency (261)	Percentage (%)
Age group		
<20	20	7.7
20-29	111	42.5
30-39	48	18.4
40-49	34	13
50-59	21	8
60 and above	27	10.3
Mean±SD	34.33±16.13	
Sex		
Male	118	45.2
Female	143	54.8
Marital Status		
Single	122	46.7
Married	130	49.8
Divorced	6	2.3
Widowed	3	1.1
Highest Education Level		
No education	20	7.7
Primary	7	2.7
Secondary	42	16.1
Tertiary	192	73.6
Occupation:		
Unemployed	26	10
Student	88	33.7
Employed (formal)	26	10
Employed (Informal)	115	44.1
Retired	6	2.3

Figure 3.2 shows the socio-economic characteristic of the participants whereby the predominant proportion of the participants 121 (46.4%) earned monthly income of ₦20,000-₦50,000.

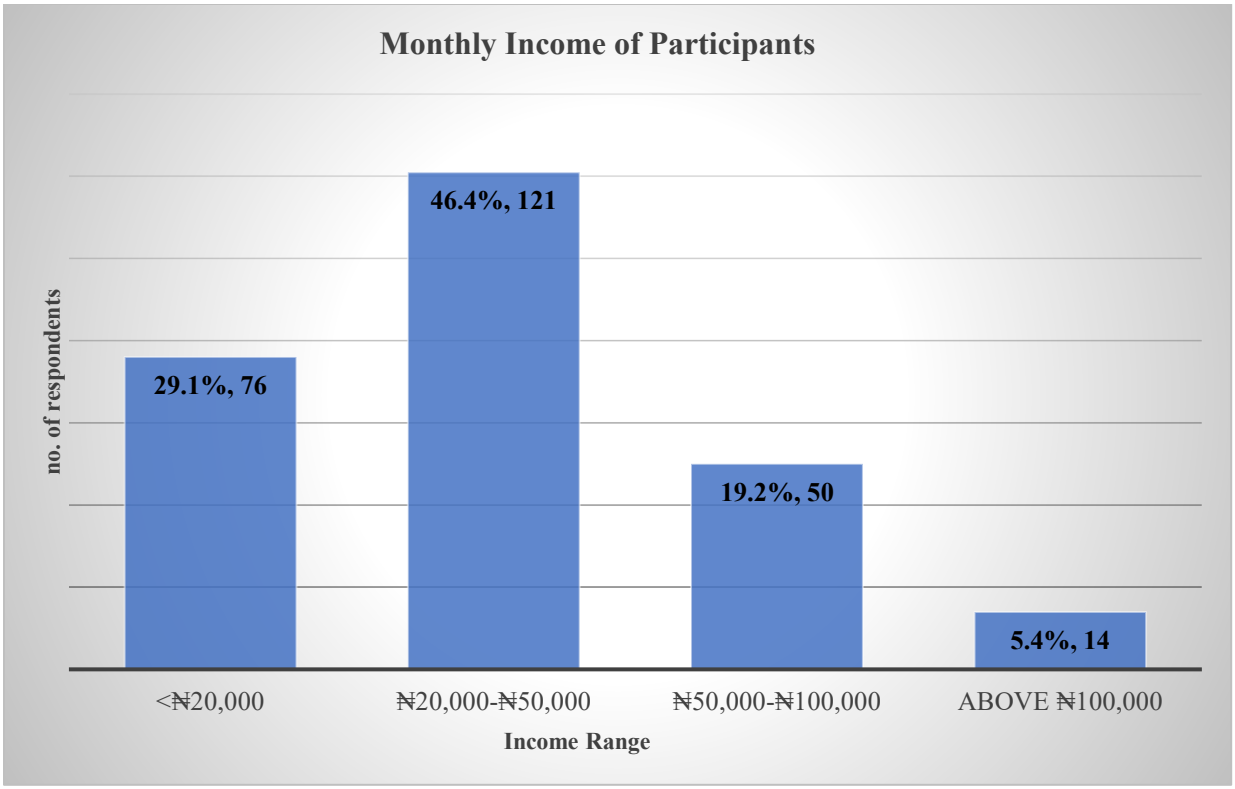


Figure 3.2: Monthly income of participants.

Table 3.2 shows the distribution of participants according to the predisposing factors for incomplete anti-tuberculosis treatment. The variables assessed include HIV statue, previous history of treatment interruption, change of treatment facility during treatment, living environments, household history of TB, contact with drug-resistant TB patients, distance to the treatment center, delays in receiving medication, experience of TB drug stock-outs and housing types.

As shown in table 3.2, participants with HIV coinfection were 30 (11.5%). Among the 149 participants who answered questions on previous treatment history, 25(16.8%) reported to have missed TB medication for more than 2 consecutive days. Also,11(7.4%) reported to have stopped TB treatment before completion and 16(10.7%) reported to have changed their TB treatment facility during treatment.

Regarding environmental and household-related factors,27(10.3%) of the participants lived in overcrowded environments, 32(12.3%) of the participants reported that someone in their household had previously had TB and 11(4.2%) participants had been in close contact with a drug-resistant TB patient.

A total number of 45(17.2%) participants lived far away (between 5-10km) from their treatment facilities. Moreover, 9(3.4%) participants experienced delays in receiving TB medication and 14(5.4%) reported TB drug stock-outs. Concerning housing type,148(56.7%) of the participants lived in rented accommodation, 71(27.2%) of them lived in their own houses and 42(16.1%) lived with family and friends.

Table 3.2: predisposing factors for incomplete anti-TB treatment.

Predisposing factors	Frequency (261)	Percentage (%)
Do you know your HIV status?		
Positive	30	11.5
Negative	215	82.4
Unknown	16	6.1
Have you ever missed TB medication for more than 2 consecutive days? (n=149)		
Yes	25	16.8
No	124	83.2
Have you ever stopped TB treatment before completion? (n=149)		
Yes	11	7.4
No	138	92.6
Did you ever change TB treatment facility during treatment? (n=149)		
Yes	16	10.7
No	133	89.3
Is your living environment overcrowded?		
Yes	27	10.3
No	234	89.7
Has anyone in your household had TB before?		
Yes	32	12.3
No	229	87.7
Have you been in close contact with a drug-resistant TB patient?		
Yes	11	4.2
No	250	95.8
Distance to treatment center		
<5km	216	82.8
5-10km	45	17.2
Have you experienced delay in receiving medication?		
Yes	9	3.4
No	252	96.6
Have you ever experienced stock-out of TB drugs?		
Yes	14	5.4

No	247	94.6
Housing Type		
Living with family/friends	42	16.1
Rented	148	56.7
Owned	71	27.2

Figure 3.3 shows the reasons given by the participants for missing their anti-TB medications for more than 2 consecutive days. The most common reason reported was forgetfulness, accounting for 8(32%) of the participants, followed by side effects with 7(28%). Other reasons included felt better (6,24%), drug unavailable (3,12%) and financial constraints (1,4%).

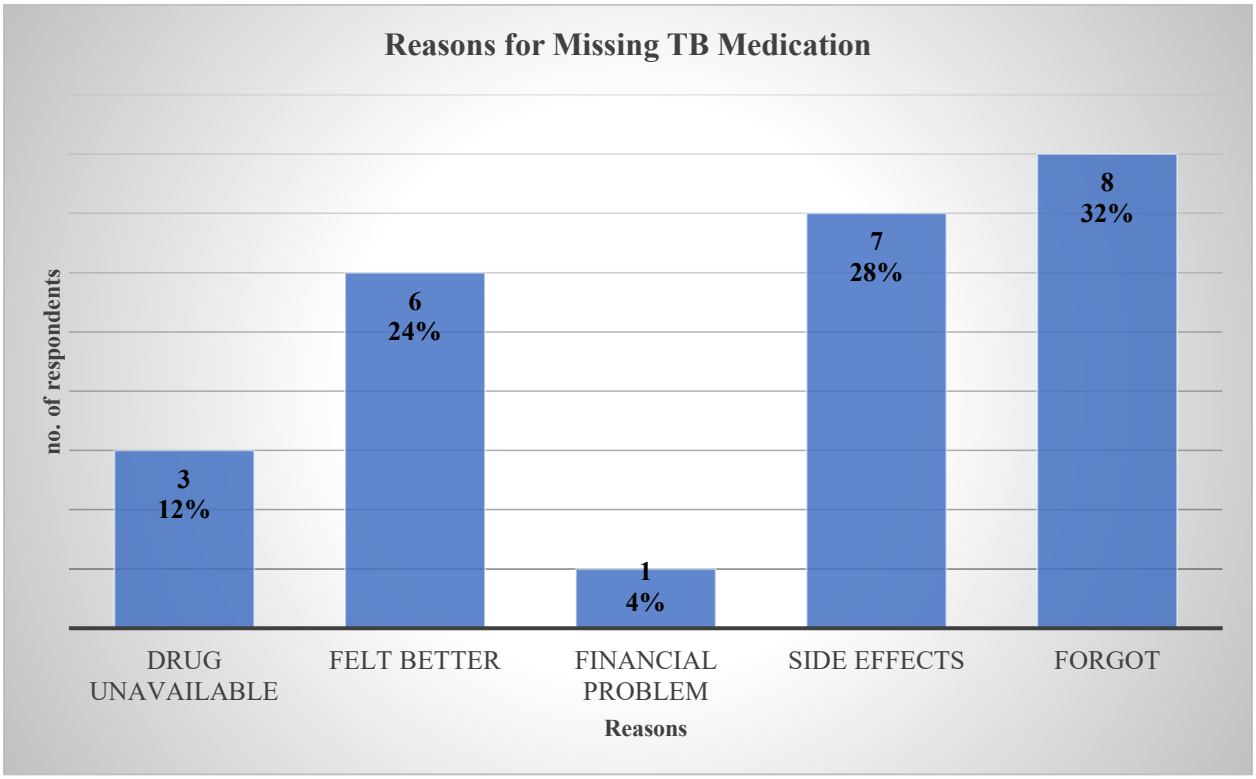


Figure 3.3: Reasons for Missing TB Medication for 2 Consecutive days.

Figure 3.4 presents the reasons reported by the participants for stopping their TB treatment before completion. The most reported reason was felt better with 8 (72.1%). Others included lack of drug access to drugs (1,9.1%) and side effects (2,18.2%).

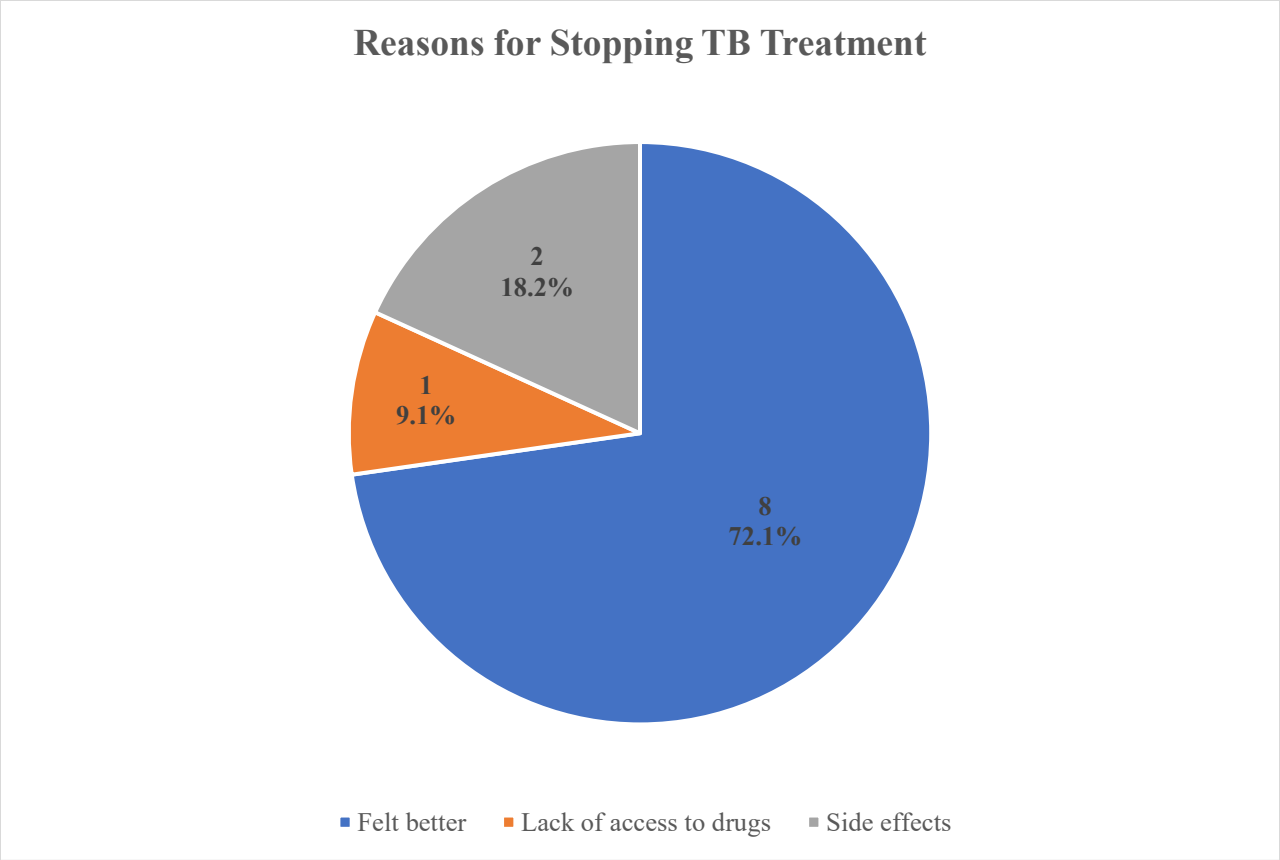


Figure 3.4: Reasons for Stopping TB Treatment Among the Patients.

Figure 3.5 illustrates the other chronic health challenges experienced by participants apart from HIV infection. The most frequently reported chronic health challenge was high blood pressure affecting 3(60%) of the participants. Other conditions identified included advanced lung disease (1,20%) and diabetes mellitus (1,20%). Overall, these findings indicate that aside from HIV, a small proportion of participants had other chronic health conditions.

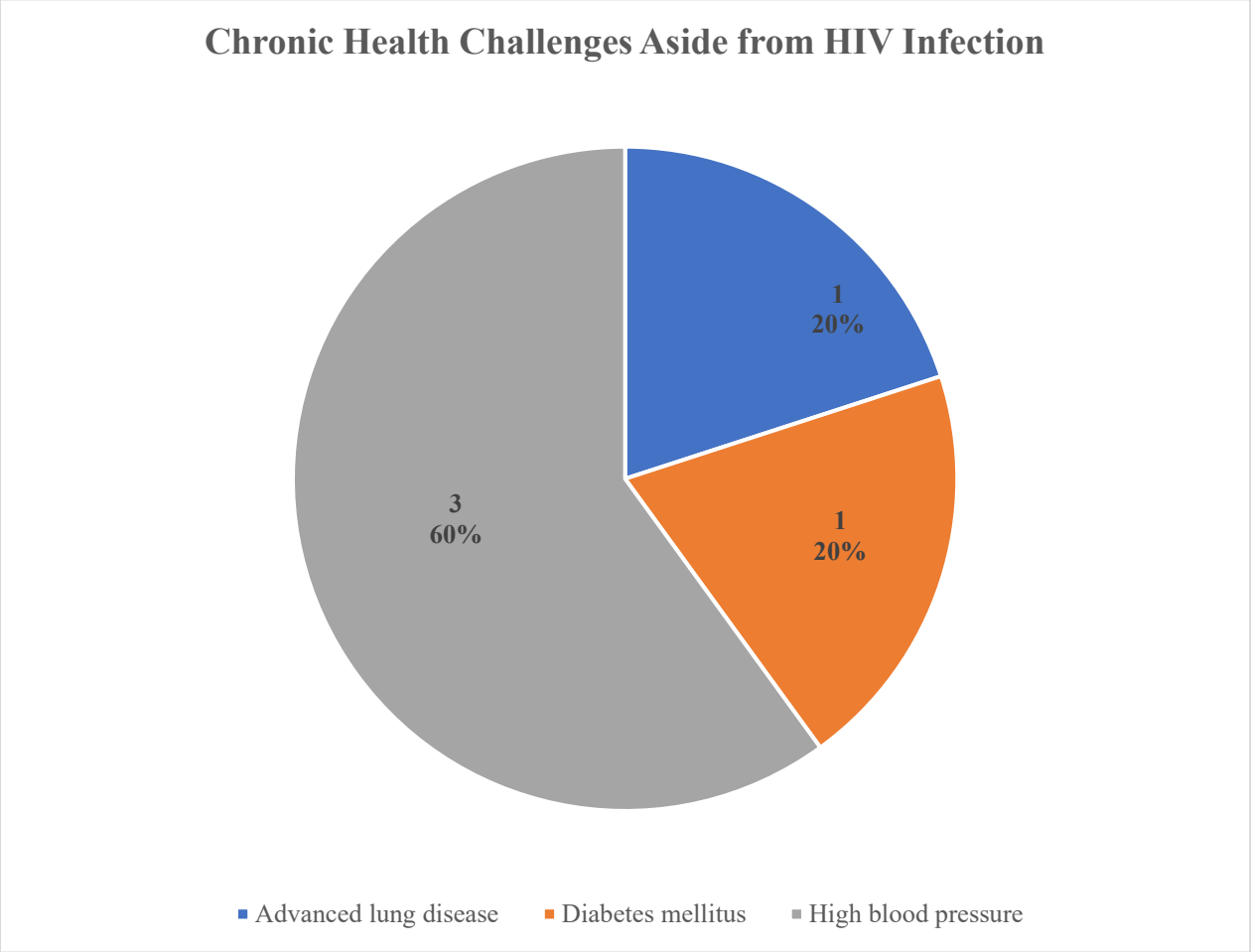


Figure 3.5: Chronic Health Challenges Aside from HIV Infection Among TB Patients.

Figure 3.6 presents the reasons participants changed their TB treatment facility during treatment. The most reported reason was relocation accounting for 11(68.8%) of the participants who changed their treatment facilities. This was followed by personal preference with 4(25%) and distance to facility with 1(6.3%).

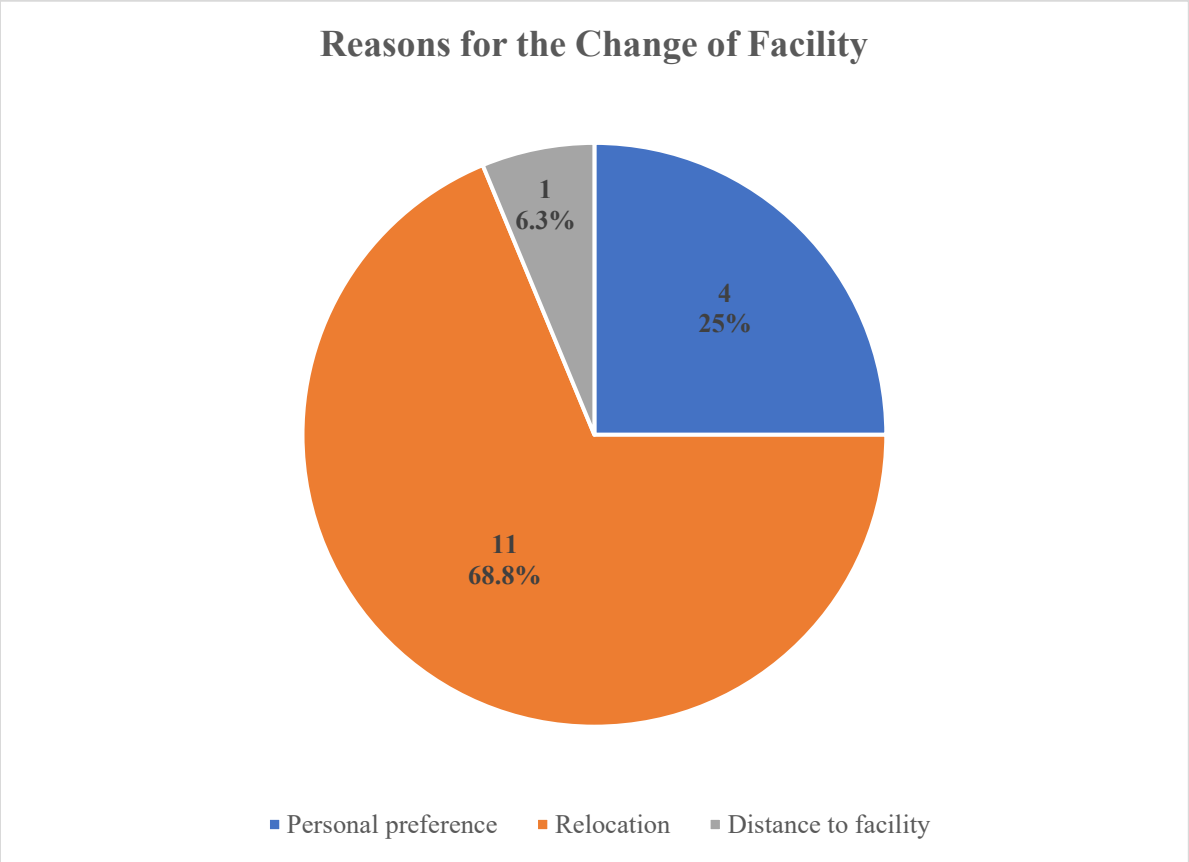


Figure 3.6: Reasons for the Change of Facility Among TB Patients.

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Table 3.3 presents the association between identified predisposing factors and drug-resistant tuberculosis (DR-TB) among the study participants. The identified predisposing factors that were significantly associated with the emergence of drug-resistant tuberculosis (DR-TB) among the 11 participants were far distance (between 5-10km) away from the treatment center ($p<0.001$), close contact with a drug-resistant TB patient ($p=0.018$), living with TB patient in household($p=0.001$) and access to free care($p=0.001$).

Table 3.3: Association between predisposing factors and drug-resistant TB

Predisposing factors	DR-TB status		X	df	p-value
	Yes	No			
Do you know your HIV status?					
Positive	2 (18.2)	28 (11.2)	0.74	2	0.69
Negative	8 (72.7)	207 (82.8)			
Unknown	1 (9.1)	15 (6.0)			
Stopped TB treatment before completion (n=149)					
Yes	2 (18.2)	9 (6.5)	2.03	1	0.155
No	9 (81.8)	129 (93.5)			
Changed TB treatment facility during treatment (n=149)					
Yes	1 (9.1)	15 (10.9)	0.034	1	0.855
No	10 (90.9)	123 (89.1)			
Have you been in close contact with a drug-resistant TB patient?					
Yes	2 (18.2)	9 (3.6)	5.55	1	**0.018**
No	9 (81.8)	241 (96.4)			
Is your living environment overcrowded?					
Yes	3 (27.3)	24 (9.6)	3.55	1	0.06
No	8 (72.7)	226 (90.4)			
Has anyone in your household had TB before?					
Yes	5 (45.5)	27 (10.8)	11.76	1	**0.001**
No	6 (54.5)	223 (89.2)			
Distance to treatment center					
<5km	4 (36.4)	212 (84.8)	17.32	1	**<0.001**
5-10km	7 (63.6)	38 (15.2)			
Have you experienced delay in receiving medication?					
Yes	1 (9.1)	8 (3.2)	1.09	1	0.295
No	10 (90.9)	242 (96.8)			
Have you ever experienced stock-out of TB drugs?					
Yes	1 (9.1)	13 (5.2)	0.31	1	0.575
No	10 (90.9)	237 (94.8)			
Can you afford to buy prescribed medications if not available at the hospital?					
Yes	0 (0)	7 (2.8)	0.316	1	0.574
No	11 (100)	243 (97.2)			
Do you have health insurance or access to free care?					
Yes	10 (90.9)	96 (38.4)	12.05	1	**0.001**

No	1 (9.1)	154 (61.6)			
Monthly Income:					
<N20,000	2 (18.2)	74 (29.6)	6.4	3	0.094
N20,000-N50,000	9 (81.2)	112 (44.8)			
N50,000-N100,000	0 (0)	50 (20.0)			
Above N100,000	0 (0)	14 (5.6)			
Household Size					
1-3.	4 (36.4)	93 (37.2)	0.033	2	0.983
4-6.	6 (54.5)	138 (55.2)			
7 and above	1 (9.1)	19 (7.6)			
Type of Residence:					
Urban	8 (72.7)	145 (58.0)	0.942	1	0.332
Rural	3 (27.3)	105 (42.0)			
Housing Type:					
Living with family/friends	1 (9.1)	41 (16.4)			
Rented	8 (72.7)	140 (56.0)			
Owned	2 (18.2)	69 (27.6)			

Table 3.4 presents the binary logistics regression analysis of factors associated with drug-resistant tuberculosis among the study participants, variables that showed significant associations in the analysis were included in the regression model to identify independent predictors of drug-resistant TB.

As presented in table 3.4, after adjusting for other variables, far distance (between 5-10km) away from the treatment center remained the only significant independent predictor of drug-resistant TB (AOR=7.509, 95% CI: 1.788-31.533, p=0.006). Participants who lived 5-10km from the treatment center had significantly higher odds of having drug-resistant TB compared to those living within 5km. Although close contact with a drug-resistant TB patient, previous household history of TB and health insurance/access to free care showed increased adjusted odds ratios, these variables were not statistically significant after adjustment.

Table 3.4: Binary logistic regression analysis of factors associated with drug-resistant TB

Predisposing factors		p-value	AOR	95% C.I.for AOR	
				Lower	Upper
In close contact with a drug-resistant TB patient?					
	Yes	0.502	2.085	0.244	17.840
	No	**Ref**			
Has anyone in your household had TB before					
	Yes	0.087	3.861	0.823	18.113
	No	**Ref**			
Distance to treatment center					
	5-10km	0.006	7.509	1.788	31.533
	<5km	**Ref**			
Have health insurance	Yes	0.065	7.504	0.883	63.741
	No	**Ref**			

CHAPTER FOUR

4.0 DISCUSSION, CONCLUSION AND RECOMMENDATION

4.1 DISCUSSION

The study titled "Assessment of Predisposing Factors to Treatment Failure Among Tuberculosis Patients Attending TB Clinics in Kwara State" aimed to determine the predisposing factors responsible for treatment failure and the emergence of drug-resistant tuberculosis among patients attending TB clinics in Kwara State. In this study, the prevalence of drug-resistant TB was 4.2%. Among the predisposing factors to TB treatment failure identified in this study include: treatment interruption, changing treatment facilities, overcrowded living conditions, previous household TB, contact with drug-resistant TB patients, delays in receiving medication or TB stock outs and far distance to the treatment facility. The identified predisposing factors associated with the emergence of DR-TB among the studied participants include far distance to the treatment facility between 5-10km ($p<0.001$), close contact with drug-resistant tuberculosis patient ($p=0.018$) and history of contact with TB patients in household ($p=0.001$).

The moderately low prevalence recorded in this study may be probably due to fewer number of facilities where the study sample was drawn predominantly from patients receiving routine, facility-based care with generally good self-reported adherence, rather than from a population based sample for treatment failure or retreatment cases, which agrees with the study conducted by Onyedum *et al.,* (2017). Moreover, drug resistance to *Mycobacterium tuberculosis* arises largely from selective pressure created by inadequate drug exposure, whether through irregular dosing, premature discontinuation, sub-therapeutic drug levels, or transmission of an already-resistant strain from infectious contact; a lower burden of interrupted treatment in this study is therefore consistent with the modest resistance prevalence observed. The implication is that while the immediate burden of resistance in this population is not alarming, even a small proportion of undetected or inadequately managed resistant cases can seed onward transmission, underscoring the continued need for universal drug-susceptibility testing rather than reliance on clinical risk profiling alone.

The study finds some of participants on treatment to have missed medication for more than two consecutive days, mainly due to forgetfulness (32%), side effects (28%), and feeling better (24%), and 7.4% discontinued treatment early, mainly on feeling better (72.7%). This finding is consistent with a cross-sectional study of Direct Observed Therapies (DOTs) patients in north-west Nigeria that similarly identified forgetfulness as a leading correlate of non-adherence (Iweama *et al.,* 2021). Similarly, in a study by Ibrahim *et al.,* (2014) conducted in Plateau state, Nigeria found that feeling well alongside transport-fare barriers were the leading reason for treatment interruption. This can be explained by the natural resolution of TB symptoms within weeks of starting effective therapy (well before bacteriological cure is achieved) commonly leads patients to underestimate their continued need for medication, while forgetfulness reflects the practical difficulty of sustaining daily dosing over a six-month or longer course, and adverse drug reactions independently discourage continued intake without adequate counselling. Hence, adherence counselling should explicitly address the gap between symptomatic improvement and bacteriological cure, side-effect management should be integrated into routine care, and reminder systems (such as phone-based prompts) could specifically target forgetfulness.

This study identifies two contact-related factors that is; having a household member with a prior history of TB and close contact with a drug-resistant TB patient, as significantly associated with DR-TB. This finding is consistent with the well-established role of primary (transmitted) resistance, in which an individual is infected de novo with an already drug-resistant strain from an infectious contact, as opposed to acquired resistance that develops as a result of non-compliance to treatment regimen (Lukoye *et al.,* 2015). The mechanism may be attributed to close-contact settings involving prolonged, high-intensity exposure to aerosolized droplet nuclei, creating ideal conditions for transmission of resistant strains where an index case exists. However, after adjustment for other variables in the logistic regression model, the two contact-related factors were not significantly associated with the emergence of drug-resistant tuberculosis. This finding may be partly explained by shared underlying factors, such as living in areas with greater treatment-access barriers or reflects the limited statistical power of the study given the small number of resistant cases rather than a true absence of effect. The implication for practice is that household contact tracing and screening of close contacts of known TB and DR-TB cases remain an important, biologically justified strategy for early case detection, even where a single study's adjusted analysis does not reach statistical significance.

Moreover, findings in this study reported far distance (between 5-10km) to the treatment center emerged as the strongest and only independent predictor of drug-resistant TB after adjustment, with patients living 5–10 km away almost eight times more likely to have DR-TB than those living within 5 km (AOR = 7.51, 95% CI: 1.79–31.53, p = 0.006). This finding aligns with a substantial body of evidence from studies conducted in Uganda, Ethiopia, and other high-burden settings showing that geographic distance to a health facility is a barrier to the timely initiation, continuity, and completion of TB treatment, even where medication itself is provided free of charge (Robsky *et al.,* 2020; Olupot *et al.,* 2021). This finding may indicate that greater travel distance increases the direct and indirect costs of care, including transport fares, lost wages, and time away from work or family responsibilities, and it complicates the directly observed therapy (DOT) model that anti-TB treatment traditionally depends on; each missed or delayed clinic visit represents an opportunity for sub-optimal drug exposure that can favour the proliferation of the resistant strains, particularly during the vulnerable early weeks of treatment. The consistency of this association even after statistical adjustment for other predisposing factors strengthens the proof that geographic access to TB treatment facility is not merely a proxy for socioeconomic disadvantage but an independent structural barrier. The implication is substantial: decentralisation of TB and DR-TB services to peripheral health facilities, community-based directly observed treatment, and the use of mobile or outreach teams for patients living beyond a defined distance threshold could meaningfully reduce the risk of resistance amplification in this population.

The findings in this study revealed that most of the participants were young adults of age range 20-29 years, predominantly female, married and had attained tertiary education. Most participants earned less than 50,000 monthly, resided in urban areas, and lived in rented accommodation.

This finding indicates that TB is concentrated among young, economically active adults rather than older or pediatric age groups. This is consistent with a study in Southeastern Nigeria that similarly found participants within 16-35 years age group as most affected (Itah & Udofia, 2005). In addition, a comparable Nigerian study of trace element status in drug-sensitive and drug-resistant TB patients in Oyo State found the 28–37 year age band to be the most prevalent in both groups and attributed this to the greater social activity and contact exposure typical of this age group (Emorinken *et al.,* 2023). A possible explanation is that young adults have the widest social and occupational mixing radius (school, work, markets, transport) which increases the number and

diversity of contacts through which *Mycobacterium tuberculosis* can be transmitted, while this age group is also more likely to be economically active and thus more exposed to crowded work and transit environments. Therefore, TB screening and case-finding activities should be concentrated in settings that reach young adults specifically, including workplaces, markets, and tertiary institutions, rather than relying solely on general facility-based screening.

The study also shows that females are more predominantly affected than males. This agrees with a study of TB/HIV co-infected patients in Anambra State, Nigeria that reported more females to being affected than males (Okeke *et al.,* 2024). The observed pattern may be attributed to women's generally higher rates of clinic attendance and health-seeking behaviour that is women attend health facilities more readily and are therefore more likely to be diagnosed and captured in facility-based samples, rather than necessarily having higher true incidence. Based on this finding, gender-sensitive case-finding is needed.

This study also reveals that low-income earners (<₦50,000) are prone to tuberculosis, despite mostly urban residence and good access to water and toilet facilities. This implies that low income, rather than infrastructural deprivation, appears to be the more consistent marker of vulnerability in this population, since access to basic amenities was otherwise adequate. Similar finding was reported by Wang *et al.,* (2021) in a study in which a composite socioeconomic status (SES) index was used to find out that the majority of TB participants fell into the low-SES band (driven mainly by low monthly income), and concluded that a patient's financial standing is a crucial predictor of TB and disease outcome. A possible explanation is that low-income patients have less capacity to absorb the indirect costs of illness (transport, lost wages, food, caregiving) and are concurrently more likely to experience the malnutrition and crowded living conditions that increase both susceptibility to infection and the risk of poor treatment outcomes. This suggests the need for an Income-targeted social protection, (such as cash transfers or transport subsidies) for TB patients in the lowest income brackets, which is likely to have a greater impact on outcomes in this population than infrastructure-focused interventions alone.

From this study, a high proportion of participants had attained tertiary education. This demonstrates that TB is not only concentrated among the least educated, which suggests that formal education alone does not protect against TB risk in this setting, and that economic and occupational factors may matter more than literacy or schooling. Similarly, SES-index study found

that most TB participants clustered in the medium education band rather than the lowest and noted that occupation and educational attainment shape the course and outcome of disease more than they shape susceptibility itself (Iman *et al.,* 2021). The observed pattern reflects that education may influence health-seeking behaviour and care-pathway choice (e.g. preference for private providers) more than it influences underlying biological susceptibility to infection, which helps explain why a well-educated subject can still carry a substantial TB burden. Based on this finding, TB awareness campaigns should not assume that educated populations are self-sufficient in navigating care and guidance on where to access free or subsidized TB services may still be needed even among literate, tertiary-educated patients.

In this study, about half of the studied participants were married. Similar findings were reported by Okeke *et al.,* (2024) in a study on TB/HIV co-infected patients where they found that the highest coinfection rates were among married patients. Also, the trace-element study found that married participants were the majority in both drug-resistant and drug-sensitive groups. This pattern was linked by the authors to the spread of active TB among people who share close daily contact, such as spouses (Long *et al.,* 2022). The likely mechanism is that single status may be associated with reduced household support and closer contact with a wider, less stable social network, while marital status among cohabiting partners can also facilitate household transmission where one partner is already infected; the net effect on any given population therefore depends on local social structure. Consequently, marital status should be used as a lens for tailoring adherence and psychosocial support (involving spouses in DOT for married patients and strengthening peer or community support for single patients).

From this study, HIV-coinfection was more common among the studied participants compared to the other forms of chronic illness (e.g. diabetes mellitus, high blood pressure etc.). This demonstrates that HIV stands out by far as the dominant co-morbidity in this population, confirming its position as the single most important biological risk factor for TB alongside socio-economic determinants, while other chronic conditions play a comparatively minor role. Similarly, an ecological analysis of Nigerian TB data found state-level HIV prevalence to be one of the strongest correlates of TB notification nationally (Daniel *et al.,* 2022) and a study of TB/HIV co-infected patients in Anambra State, Nigeria similarly documented substantial and sustained TB/HIV co-infection rates over a five-year period (Okeke *et al.,* 2024). A possible explanation is

that HIV-related immunosuppression markedly increases both the risk of progression from latent to active TB and the risk of reactivation of prior infection, an effect with no comparable counterpart among the other less prevalent chronic conditions reported in this study. From a public health perspective, TB–HIV integration should remain the primary comorbidity-focused strategy in this setting and resources devoted to comorbidity screening should be concentrated on HIV testing.

This study reveals that more than half of participants lived in rented accommodation. Similarly, the study on trace-element among TB patients found most participants stay in substandard housing which was attributed to Nigeria's broader housing deficit and its consequences on disease spread (Akande *et al.,* 2024). This is likely because renting is often associated with more frequent relocation, less control over ventilation and living conditions, and lower investment in home improvements, all of which can indirectly elevate exposure risk even where the number of occupants per room is not itself excessive. Based on this finding, housing-focused social protection (such as rent support or advocacy for minimum ventilation standards in rented housing) may be a relevant complementary intervention alongside direct TB care, particularly for patients who also report difficulty affording treatment.

From the findings in this study, most participants resided in urban areas. This may be due to the increase migration to urban areas responsible for the greater population density, mobility, and congregate settings characteristic of urban life. Similarly, Urban residence was found to independently predict catastrophic TB-related expenditure in the Ebonyi State (Ukwaja *et al.,* 2013a). Also, the national ecological analysis of TB in Nigeria similarly linked urbanization-related factors to TB notification patterns (Daniel *et al.,* 2022). A possible explanation is that urban environments concentrate people in markets, public transport, and multi-occupancy housing, increasing the frequency and density of person-to-person contact through which airborne transmission occurs, while urban cost-of-living pressures can simultaneously deepen the financial burden of illness. Therefore, urban-focused active case-finding, (e.g. in markets, motor parks, and other congregate settings) is likely to yield a higher case detection rate per unit effort than a purely facility-based approach in this setting.

4.1.1 STRENGTHS AND LIMITATIONS OF THE STUDY

4.1.1.1 STRENGTHS

This study has several notable strengths. It employed a reasonably large sample size of 261 patients drawn from routine care, allowing for meaningful bivariate and multivariable analysis of predisposing factors. The use of binary logistic regression to adjust for potential factors strengthens the internal validity of the findings on independent predictors of drug resistance. The study also captured a broad range of potential predisposing factors spanning clinical history, health-system access, socioeconomic status, and housing conditions, providing a comprehensive picture of the multifactorial determinants of incomplete treatment and drug resistance in this population. Additionally, the inclusion of self-reported reasons for missed doses and treatment discontinuation adds behavioural depth that purely quantitative associations cannot capture, offering actionable detail for programme design.

4.1.1.2 LIMITATIONS

Several limitations noticed when interpreting these findings include:

- Cross-sectional design means that temporal and causal relationships between the variables examined, and TB or DR-TB status cannot be established; the study can identify associations but not confirm which factor preceded which.
- The small absolute number of drug-resistant cases (n = 11) limited the statistical power of the multivariable model, producing wide confidence intervals around several adjusted odds ratios and likely explaining why some bivariate associations lost significance after adjustment.
- Key exposures, including HIV status, treatment adherence history, and reasons for missed doses, were based on self-report rather than laboratory or pharmacy-refill verification, introducing the possibility of recall and social-desirability bias, particularly around sensitive topics such as treatment discontinuation.
- The limited-facility nature typical of studies of this kind restricts the generalisability of the findings to other geographic or healthcare contexts with different service configurations.
- Finally, distance to the treatment center was assessed using broad categorical bands rather than precise geospatial measurement.

4.2 CONCLUSION

This study sets out to examine the predisposing factors for incomplete anti-TB treatment and the determinants of drug-resistant tuberculosis among 261 patients. The findings show that while overall drug resistance prevalence in this population was relatively favourable, at 4.2%, specific structural and epidemiological factors meaningfully increased the risk of resistant disease. Distance to the treatment center emerged as the single strongest and only independent predictor of drug-resistant TB after statistical adjustment, while contact with a drug-resistant TB patient and household TB history were each significantly associated with resistance on bivariate analysis, pointing to the intertwined roles of transmission dynamics, health-system access, and health-seeking behaviour in shaping resistance risk. Forgetfulness, drug side effects, and a premature sense of cure were identified as the principal behavioural drivers of missed doses and treatment discontinuation, reaffirming patterns long documented elsewhere in the literature.

Scientifically, this study contributes locally generated, facility-level evidence on the determinants of drug-resistant TB in a setting where such data remains limited, and it adds to a growing body of evidence identifying geographic access, rather than household socioeconomic status alone, as a key structural driver of resistance risk. It further contributes detailed

evidence on the complex, potentially non-causal relationship between health insurance access and resistance status, an association that merits closer investigation in future longitudinal research. Practically, the findings offer TB control programmes actionable, evidence-based targets for intervention, namely, reducing geographic barriers to treatment access, strengthening contact tracing and household screening, and addressing the specific behavioural drivers of non-adherence through structured patient education and reminder systems.

Ultimately, this study reaffirms that drug-resistant tuberculosis is not simply a biological accident but a preventable consequence of gaps in access, continuity, and support within the systems designed to deliver TB care. Closing these gaps, particularly the geographic and behavioural barriers identified here, represents an achievable and necessary step toward reducing the future burden of drug-resistant tuberculosis in this population and in comparable settings.

4.3 RECOMMENDATION

Based on the findings of this study, the following recommendations are made:

1. **Decentralise TB and DR-TB services:** National and state TB control programmes should extend directly observed treatment, drug collection points, and follow-up services closer to patients living beyond 5 km from the nearest treatment center, given that distance was the strongest independent predictor of drug resistance identified in this study.
2. **Geospatial analysis:** using precise travel-time or GPS-based distance measures, rather than categorical bands, would help to define actionable distance thresholds for decentralising services.
3. **Longitudinal or cohort studies** are recommended to establish the temporal sequence and causal pathways linking the socio-demographic and behavioural variables examined here to the development of active TB and drug resistance.
4. **Multi-facility and multi-state studies** are recommended to determine whether the gender, marital status, and residence patterns observed here are specific to this study population or generalisable more broadly.
5. **Expand the evidence base through larger, multi-site studies:** Future studies should pool data across multiple facilities to achieve the statistical power needed to more definitively characterize independent predictors of drug-resistant TB, particularly given the small number of resistant cases available for analysis in this study.
6. **Deploy community-based support and reminder systems:** Treatment supporters, mobile-phone-based reminders, and community health worker follow-up should be integrated into routine care to address forgetfulness, the single most common reason for missed doses identified in this study.
7. **Enhance patient education at treatment initiation:** Structured counselling at treatment initiation should explicitly address the expected timeline of symptom improvement and the likely side-effect profile of anti-TB drugs, to reduce premature discontinuation driven by a false sense of cure or by unmanaged side effects.
8. **Investigate the insurance-resistance relationship further:** TB programmes and researchers should conduct longitudinal studies to clarify whether health insurance access is a marker of pre-existing disease severity and healthcare-seeking behaviour or a

modifiable risk factor, before drawing programmatic conclusions from the association observed here.

9. **Integrate post-treatment clinical follow-up:** TB programmes should extend follow-up beyond treatment completion to screen for chronic sequelae such as residual lung impairment and diabetes mellitus among TB survivors, given the chronic health changes reported by a subset of participants in this study.
10. **Maintain universal drug-susceptibility testing:** Given that resistance was not reliably predicted by clinical risk profiling alone (e.g. the absence of an association with HIV status in this study), universal drug-susceptibility testing should continue to be prioritised over risk-based selective testing wherever laboratory capacity allows.
11. **Strengthen contact tracing and household screening:** Active screening should be intensified for household members and close contacts of known TB and drug-resistant TB patients, using symptom screening and, where feasible, molecular diagnostic testing, to enable early detection of transmitted resistant disease.

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APPENDIX I

Questionnaire on Factors Associated with Emergence of DR-TB in Kwara State

Instructions:

This questionnaire is for research purposes only.

All answers will be kept confidential.

Tick (✓) or write your answer as appropriate.

PARTICIPANT ID: …….

PHONE NUMBER: _____

SECTIONA: Socio-Demographic & Socio-Economic Characteristics

1. Age: …… years

2. Sex: ☐Male ☐Female

3.Marital Status: ☐Single ☐Married ☐Divorced ☐Widowed

4.Highest Education Level: ☐ No formal education ☐Primary ☐Secondary ☐Tertiary

5.Occupation: ☐Employed (formal) ☐Employed (informal) ☐Unemployed ☐Student ☐Retired

6.Monthly Income: ☐<₦20,000☐₦20,000–₦50,000☐₦50,000–₦100,000☐>₦100,000

7.Household Size (number of people living together): ☐1–3 ☐4–6 ☐≥7

8.TypeofResidence: ☐Urban ☐Rural

9.HousingType: ☐Owned ☐Rented ☐Living with family/friends

10.Access to clean water: ☐Yes ☐No

11.Access to Toilet Facility: ☐Yes ☐No

SECTION B: Clinical History

12.Haveyou been treated for TB before? ☐Yes ☐No

13.Ifyes, how many times? ☐Once ☐Twice ☐More than twice

14.Duration of previous TB treatment: ☐< 6 months ☐6–12 months☐> 12 months

15.Do you know your HIV status? ☐Positive ☐Negative ☐Unknown

16.Do you have any other chronic illness? ☐Yes ☐No

17.Ifyes, ☐Diabetes mellitus ☐Advanced lung disease ☐Others: _____

SECTION C: Treatment Adherence & Drug Use

18.Areyou currently taking TB medication as prescribed? ☐Yes ☐No

19.When do you take your medication? ☐Prescribed time ☐Convenient time ☐Irregular timing

20. How do you take your prescribed TB medication dose? ☐Full dose as prescribed ☐Sometimes reduce the dose ☐Sometimes skip doses ☐Not sure of correct dose

21.Haveyou ever missed TB medication for more than 2 consecutive days? ☐Yes ☐No

22.Ifyes, reason: ☐Side effect ☐Forgot ☐Financial problems ☐Felt better ☐Drug unavailable ☐Others: _____

23.Haveyou ever stopped TB treatment before completion? ☐Yes ☐No

24.Ifyes, reason: ☐Side effects ☐Felt better ☐Financial constraints ☐Lack of access to drugs ☐Others: _____

25.Wereyou placed on Directly Observed Therapy (DOT)? ☐Yes ☐No

26.Did you ever change TB treatment facility during treatment? ☐Yes ☐No

27.Ifyes, reason: ☐Relocation ☐Distance to facility ☐Poor service ☐Personal preference ☐Others: _____

28.Haveyou ever used herbal or alternative medicine for TB? ☐Yes ☐No

SECTION D: Lifestyle& Risk Factors

29.Do you smoke? ☐Yes ☐No.

30.Ifyes, how often: ☐Daily ☐Occasionally ☐Never

31.Do you consume alcohol? ☐Yes ☐No

32.Ifyes, how often: ☐Daily ☐Occasionally ☐Never

33.Haveyou been in close contact with a drug-resistant TB patient? ☐Yes ☐No

34.Haveyou ever been incarcerated? ☐Yes ☐No

SECTION E: Household & Environmental Factors

35.Isyour living environment overcrowded? ☐Yes ☐No

36.Has anyone in your household had TB before? ☐Yes ☐No

SECTION F: Health System &Access Factors

37.Distanceto treatment center: ☐far ☐near

38.Haveyou experienced delay in receiving medication? ☐Yes ☐No

39.Haveyou ever experienced stock-out of TB drugs? ☐Yes ☐No

40.Can you afford to buy prescribed medications if not available at the hospital? ☐Yes ☐No

41.Do you have health insurance or access to free care? ☐Yes ☐No

Outcome Variable (from records)

DR-TB status: ☐Yes ☐No